

第9回 一般の周期関数のフーリエ級数 (1)

今回の内容

一般の周期関数のフーリエ級数 (1)

第3章 フーリエ解析 §1

定義：フーリエ展開（周期 $T = 2l$ ）

$$f(t) = c_0 + \sum_{n=1}^{\infty} (a_n \cos(n\pi t/l) + b_n \sin(n\pi t/l))$$

フーリエ係数：

$$c_0 = (1/2l) \int_{-l}^l f(t) dt$$

$$a_n = (1/l) \int_{-l}^l f(t) \cos(n\pi t/l) dt (n = 1, 2, \dots)$$

$$b_n = (1/l) \int_{-l}^l f(t) \sin(n\pi t/l) dt (n = 1, 2, \dots)$$

例題 3 (p. 83)

問題

$f(x) = |x|(-1 \leq x < 1), f(x+2) = f(x)$ のフーリエ展開を求めよ

例題3 解答

$f(x)$ は偶関数なので $b_n = 0$, フーリエ余弦展開を求めればよい

$$c_0 = (1/2) \int_{-1}^1 |x| dx = 1/2$$

$$a_n = 2 \int_0^1 x \cos(n\pi x) dx = -2(1 - (-1)^n)/(n^2\pi^2)$$

答: $f(x) = 1/2 - (4/\pi^2)(\cos \pi x + \cos(3\pi x)/9 + \cos(5\pi x)/25 + \dots)$

$N=2, N=5$ での収束グラフあり (p.83)

自分で解いてみよう

問3 次の関数のフーリエ級数を求めよ

$$(1) f(x) = \begin{cases} 1 & (-1 \leq x < 0) \\ 0 & (0 \leq x < 1) \end{cases}, f(x+2) = f(x)$$

$$(2) g(x) = \begin{cases} 2 & (-2 \leq x < 0) \\ 4 & (0 \leq x < 2) \end{cases}, g(x+4) = g(x)$$

$$(3) h(x) = \begin{cases} 2x+1 & (-1/2 \leq x < 0) \\ 0 & (0 \leq x < 1/2) \end{cases}, h(x+1) = h(x)$$

問3 解答

$$(1) l = 1 : c_0 = 1/2, a_n = \int_{-1}^0 \cos(n\pi x) dx = 0, b_n = \int_{-1}^0 \sin(n\pi x) dx = ((-1)^n - 1)/(n\pi)$$

$$(2) l = 2 : c_0 = 3, a_n = 0, b_n = (1/2)[2 \int_{-2}^0 \sin(n\pi x/2) dx + 4 \int_0^2 \sin(n\pi x/2) dx]$$
$$b_n = 2(1 - (-1)^n)/(n\pi)$$

$$(3) l = 1/2 : c_0 = 1/4, a_n = (1 - (-1)^n)/(n^2\pi^2), b_n = -1/(n\pi)$$

答 : (1) $f(x) = 1/2 - (2/\pi)(\sin \pi x + \sin(3\pi x)/3 + \sin(5\pi x)/5 + \dots)$
(2) $g(x) = 3 + (4/\pi)(\sin(\pi x/2) + \sin(3\pi x/2)/3 + \sin(5\pi x/2)/5 + \dots)$
(3) $h(x) = 1/4 + \sum_{n=1}^{\infty} [(1 - (-1)^n)/(n^2\pi^2) \cos(2n\pi x) - (1/(n\pi)) \sin(2n\pi x)]$